Arman Aydin

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Education

Stanford University — Stanford, CA
Master of Science in Computer Science — *Jan 2024 – Jun 2025*
Bachelor of Science in Computer Science — *Sep 2020 – Jun 2024*

Experience

Box Inc. — Software Engineer II
Python, PyTorch, Docker, Kubernetes, Kedro, Prefect, Braintrust — *Jul 2025 – Present*

- Architected synthetic data generation pipelines using Kedro, modeling open-source datasets as graphs; designed graph-traversal based algorithms (modeled on the MuSiQue framework) to programmatically construct complex, multi-hop reasoning benchmarks for enterprise-specific contexts.
- Designed and deployed an online grading pipeline modeled after Anthropic's CLIO pipeline for emergent hierarchical clustering of user activity; for offline grading, utilized Braintrust for trajectory tracing and integrated automated stability metrics using semantic entropy and self-consistency variance.
- Engineered a high-throughput automated evaluation infrastructure integrated directly into the CI/CD pipeline; orchestrated containerized batch workloads to execute 10,000+ daily test cases, enforcing empirical quality baselines critical to the General Availability (GA) launch of the Box Agents platform (scaling to 1M+ users).

Stanford Vision and Learning Lab (SVL) — Research Assistant
NVIDIA IsaacSim, PyTorch, Autodesk 3ds Max, Dask, DVC — *Mar 2023 – Sep 2024*

- Contributed on the convex mesh data pipeline for BEHAVIOR-1K, a benchmark for evaluating Embodied AI solutions through interactive household activities with physics simulations supervised by Dr. Fei-Fei Li.
- Scripting in Autodesk 3ds Max, integrating CoACD performance metrics, where achieved up to 39% speed-up in internal sanity check scripts for integrity checks of physics simulations.
- Developed teleoperation features for OmniGibson, Stanford-developed robot/environment simulation platform built on top of NVIDIA's IsaacSim, designing a VR-based trajectory collection pipeline to capture high-fidelity human demonstrations for transfer learning and offline reinforcement learning (RL) frameworks.

Stanford NLP Group — CURIS Research Assistant
PyTorch, TensorFlow, Numpy, SQL, Bash Script, JavaScript, Pytest — *Jun 2022 – Sep 2022*

- Implemented Named Entity Recognition (NER) language models for low-resource languages Kazakh (95% F1-score) and Uyghur (77% F1-score) by leveraging cross-lingual syntactic similarities with high-resource languages like Turkish through transfer learning (supervised by Dr. John Bauer and Dr. Chris Manning).
- Enhanced the Stanford-developed Stanza natural language analysis package, achieving an 11% increase in parsing speed for nested NER tags, and added generalized transliteration across scripts and word span correction features.

Publications & Selected Projects

Publications

- BEHAVIOR Vision Suite: Customizable Dataset Generation via Simulation (Presented in CVPR 2024)
- BEHAVIOR-1K: A Benchmark for Embodied AI with 1,000 Everyday Activities and Realistic Simulation (Presented in CoRL 2023)

vllm-lite: Lightweight vLLM Inference Engine | *PyTorch, Triton, PagedAttention, Continuous Batching*

- Built a high-performance LLM inference engine featuring a continuous batching scheduler with recomputation preemption and a custom, fused Triton PagedAttention decode kernel to dynamically manage non-contiguous KV cache blocks.
- Integrated FlashAttention for accelerated prefill computation and consolidated concurrent decodes into unified forward passes, achieving a 4.25x boost in generation throughput and a 95% reduction in time-to-first-token (TTFT) latency.

MixKVQ: Mixed-Precision KV Cache | *Triton, PyTorch, INT4/INT8 Bit-Packing*

- Implemented the MixKVQ architecture by building a calibration pipeline to extract outlier scales, coupled with a custom Triton packing kernel that routes critical outliers to BF16 while quantizing non-outliers to INT4/INT8, achieving a 19.5x execution speedup.
- Programmed a dequantizing FlashAttention kernel that executes on-the-fly dequantization directly in SRAM, reducing overall KV cache VRAM footprints by 40% (and isolated key caches by 64%) while matching full-precision logits to a 1e-2 tolerance.

Distributed Goal-Conditioned RL Pipeline for OGBench | *Ray, PyTorch, Multi-GPU Orchestration, Simulation*

- Engineered a distributed training and rollout pipeline for goal-conditioned RL agents; orchestrated CPU-bound simulation pools via Ray remote actors and synchronized parameters using PyTorch Distributed Data Parallel (DDP) with centralized batch broadcasting.
- Eliminated CPU-GPU bottlenecks to boost rollout throughput 30x (130 to 4,500 steps/s), utilizing Ray's shared memory Plasma store for zero-copy weight updates and FP16 mixed precision (AMP) to accelerate optimization.

Technical Skills

Languages: Python, C, C++, Rust, JavaScript, TypeScript, SQL, Julia
Frameworks: PyTorch, Triton, Node.js, Docker, Kubernetes, Scikit-learn, HuggingFace, vLLM, AWS